

8.16.20 Time-Dependent Pump Velocity Control, CCC6100

This card is required only if W5 of card CCC0301 is 0. The velocity table, if present, has priority in setting the pump velocity over the pump trip, the pump motor torque data, and the torque-inertia equation.

W1(I) Trip number. If the trip number is 0, the pump velocity is always computed from this table. If the trip number is nonzero, the trip determines which table is to be used. If the trip is off, the pump velocity is set from the trip, the pump motor torque data, and the torque-inertia equation as if this table had not been entered. If the trip is on, the pump velocity is computed from this table. If Word 2 is missing, the search variable in the table is time and the search argument is time minus the trip time. If this word is used, it takes precedence over the trip number used in Word 6 of the CCC0301 card.

W2(A) Alphanumeric part of variable request code. This quantity is optional. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. TIME can be selected, but the trip time is not subtracted from the advancement time.

W3(I) Numeric part of variable request code. This is assumed to be 0 if missing.

8.16.21 Time-Dependent Pump Velocity, CCC6101-6199

These cards are required only if W5 of card CCC0301 is 0.

W1(R) Search variable. Units depend on the quantity selected for the search variable.

W2(R) Pump velocity (rad/s, rev/min).

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs. Time values must be in increasing order.

8.17 Circulator Component, CIRCLTR

A circulator component is indicated by CIRCLTR for Word 2 on card CCC0000. A circulator consists of one volume and two junctions, one attached to each end of the volume. For major edits, minor edits, and plot variables, the volume in the circulator component is numbered as CCC010000. The circulatorcirculator junctions are numbered CCC010000 for the inlet junction and CCC020000 for the outlet junction.

8.17.1 Circulator Volume Geometry, CCC0101-0107

This card (or cards) is required for a circulator component. The seven words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R)	Area (m ² , ft ²).
W2(R)	Length (m, ft).
W3(R)	Volume (m ³ , ft ³). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the area times the length within a relative error of 0.000001.
W4(R)	Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.
W5(R)	Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward direction, i.e., the outlet is at a higher elevation than the inlet. This angle is used in the interphase drag calculation.
W6(R)	Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be equal to or less than the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. For this component, this Word 6 is not compared to the length (Word 2) to decide if the horizontal or vertical flow regime is used. Rather, the circulator flow regime map is used.
W7(I)	Volume control flags. This word has the format <u>tlpvbfe</u> . It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. The circulator component forces all volume flags except for the e digit, and y- and z-coordinate flags are not read. The effective format is <u>000000e</u> .
e-flag	The equilibrium digit <u>e</u> specifies if nonequilibrium or equilibrium is to be used; <u>e</u> = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and <u>e</u> = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

8.17.2 Circulator Inlet Junction, CCC0108

This card is required for a circulator component.

- W1(I) Volume code of connecting volume on inlet side. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, only the number N equal to 1 or 2 is allowed.
- W2(R) Area (m^2 , ft^2). If 0, the area is set to the minimum area of adjacent volumes. If an abrupt area change, the area must be equal to or less than the minimum of the adjacent volume areas. If a smooth area change, no restrictions exist.
- W3(R) Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.16.6). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W5. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W2. If one of the abrupt area change options is selected ($a = 1$ or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.
- W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.16.6). See information for the previous word regarding interpretation and use of this loss coefficient.
- W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the circulator inlet junction, the junction control flag is restricted to the format 00f0cah0.
- f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.
- c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied.

The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in the circulator component. Problem renodalization should be used to circumvent this limitation should it prove to be significant.

For the Henry-Fauske critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

- a-flag** The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.
- h-flag** The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

8.17.3 Circulator Outlet Junction, CCC0109

This card is required for a circulator component. The format for this card is identical to card CCC0108 except data are for the outlet junction.

This card is required for a circulator component.

- W1(I)** Volume code of connecting volume on outlet side. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.
- W2(R)** Area (m^2 , ft^2). If 0, the area is set to the minimum area of adjacent volumes. If an abrupt area change, the area must be equal to or less than the minimum of the adjacent volume areas. If a smooth area change, no restrictions exist.

W3(R)	Reynolds number independent forward flow energy loss coefficient, A_F . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.16.6). The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected in W5. If the smooth area change option is selected ($a = 0$), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W2. If one of the abrupt area change options is selected ($a = 1$ or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the $a = 1$ option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.
W4(R)	Reynolds number independent reverse flow energy loss coefficient, A_R . This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.16.6). See information for the previous word regarding interpretation and use of this loss coefficient.
W5(I)	Junction control flags. This word has the format <u>jefvcahs</u> . It is not necessary to input leading zeros. For the circulator outlet junction, the junction control flag is restricted to the format <u>00f0cah0</u> .
f-flag	The digit <u>f</u> specifies CCFL options. <u>f</u> = 0 means that the CCFL model will not be applied, and <u>f</u> = 1 means that the CCFL model will be applied.
c-flag	The digit <u>c</u> specifies choking options. <u>c</u> = 0 means that the choking model will be applied, and <u>c</u> = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in the circulator component. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the Henry-Fauske critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.
a-flag	The digit <u>a</u> specifies area change options. <u>a</u> = 0 means a smooth area change and only the user supplied K_{loss} is applied. <u>a</u> = 1 means full abrupt area change model and the code

calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. $\underline{a} = 2$ means a partial abrupt area change model and it is the same as $\underline{a} = 1$ except no code calculated forward and reverse contraction/expansion K_{loss} is applied.

h-flag The digit $\underline{h}$ specifies nonhomogeneous or homogeneous. $\underline{h} = 0$ specifies the nonhomogeneous (two-velocity momentum equations) option; $\underline{h} = 1$ or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option ($\underline{h} = 1$ or 2), the major edit printout will show a 1.

8.17.4 Circulator Inlet Junction Form Loss Data, CCC0112

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC0108 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 3 and 4 of card CCC0108. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R .

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

8.17.5 Circulator Outlet Junction Form Loss Data, CCC0113

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC0109 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F \text{Re}^{-C_F}$$

$$K_R = A_R + B_R \text{Re}^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 3 and 4 of card CCC0109. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R ; the same information applies for B_F and B_R . The format of this card is identical to card CCC0112 except data are for the outlet junction.

W1(R) $B_F (\geq 0)$. This quantity must be greater than or equal to 0.

W2(R) $C_F (\geq 0)$. This quantity must be greater than or equal to 0.

W3(R) $B_R (\geq 0)$. This quantity must be greater than or equal to 0.

W4(R) $C_R (\geq 0)$. This quantity must be greater than or equal to 0.

8.17.6 Circulator Volume Initial Conditions, CCC0200

This card is required for a circulator component.

W1(I) Control word. This word has the format $\underline{\epsilon}\text{bt}$. It is not necessary to input leading zeros.

e-flag The digit $\underline{\epsilon}$ specifies the fluid, where $\underline{\epsilon} = 0$ is the default fluid. The value for $\underline{\epsilon} > 0$ corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., $\underline{\epsilon} = 1$ specifies H_2O , $\underline{\epsilon} = 2$ specifies D_2O , etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default $\underline{\epsilon} = 0$, then H_2O is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit $\underline{b}$ specifies whether boron is present. $\underline{b} = 0$ specifies that the volume liquid does not contain boron; $\underline{b} = 1$ specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit $\underline{t}$ specifies how the following words are to be used to determine the initial thermodynamic state. $\underline{t} = 0-3$ specifies one component (steam/water). Entering $\underline{t} = 4-6$ allows the specification of two components (steam/water and noncondensable gas).

With options $\underline{t}$ equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g, α_g] If $\underline{t} = 0$, the next four words are interpreted as pressure (Pa, lb_f/in²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If $\underline{t} = 1$, the next two words are interpreted as temperature (K, °F) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If $\underline{t} = 2$, the next two words are interpreted as pressure (Pa, lb_f/in²) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If $\underline{t} = 3$, the next two words are interpreted as pressure (Pa, lb_f/in²) and temperature (K, °F) in equilibrium condition. Enter boron in W4 if needed.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0\text{E-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0 \text{ E-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g / (M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If $\underline{t} = 4$, the next three words are interpreted as pressure (Pa, lb_f/in.²), temperature (K, °F), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0 , saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensables (dry noncondensable)

with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T,x_s,x_n] If $\underline{t} = 5$, the next three words are interpreted as steam saturation temperature (K, °F), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

[P,U_f,U_g,α_g,x_n] If $\underline{t} = 6$, the next five words are interpreted as pressure (Pa, lb_f/in.²), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing ($\underline{t} = 0$). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

8.17.7 Circulator Inlet Junction Initial Conditions, CCC0201

This card is required for a circulator component.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flow rates.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W4(R) Initial interface velocity (m/s, ft/s). Enter 0.

8.17.8 Circulator Outlet Junction Initial Conditions, CCC0202

This card is required for a circulator component. This card is similar to card CCC0201 except that data are for the outlet junction.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flow rates.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W4(R) Initial interface velocity (m/s, ft/s). Enter 0.

8.17.9 Circulator Index and Option, CCC0301

This card is required for a circulator component.

W1(I) circulator motor torque table index. If -1, no table is used. If 0, a table is entered for this component. If nonzero, use the table from the component with this number.

W2(I) Time-dependent circulator velocity index. If -1, no time-dependent circulator rotational velocity table is used and the circulator velocity is always determined by the torque-inertia equation. If 0, a table is entered with this component. If nonzero, the table from the circulator component with this number is used. A circulator velocity table cannot be used when the circulator is connected to a shaft control component.

W3(I) circulator trip number. When the trip is off, electrical power is supplied to the circulator motor; when the trip is on, electrical power is disconnected from the circulator motor. The circulator velocity depends on the circulator velocity table and associated trip, the circulator motor torque data, and this trip. If the circulator velocity table is being used, the circulator velocity is always computed from that table. If the circulator velocity table is not being used, the circulator velocity depends on the circulator motor torque data and this trip. If the trip is off and no circulator motor torque data are present, the circulator velocity is the same as for the previous time step. This will be the initial circulator velocity if the circulator trip has never been set. Usually the circulator trip is a latched trip, but that is not necessary. If the trip is off and a circulator motor torque table is present, the circulator velocity is given by the torque-inertia equation where the net torque is given by the circulator motor torque data and the homologous torque data. If the trip is on, the torque-

inertia equation is used and the circulator motor torque is set to 0. If the circulator trip number is 0, no trip is tested and the circulator trip is assumed to always be off.

8.17.10 Circulator Description, CCC0302-0304

This card (or cards) is required for a circulator component.

W1(R)	Rated circulator velocity (rad/s, rev/min).
W2(R)	Ratio of initial circulator velocity to rated circulator velocity. Used for calculating initial circulator velocity.
W3(R)	Rated flow (m^3/s , gal/min).
W4(R)	Rated head (m, ft).
W5(R)	Rated torque ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$).
W6(R)	Moment of inertia ($\text{kg}\cdot\text{m}^2$, $\text{lb}_f\cdot\text{ft}^2$). This includes all direct coupled rotating components, including the master for a motor driven circulator.
W7(R)	Rated density (kg/m^3 , lb/ft^3). If 0, initial density is used. This is the density used to generate homologous data.
W8(R)	Rated circulator motor torque ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$). If this word is 0.0, the rated circulator motor torque is computed from the initial circulator velocity and the circulator torque that is computed from the initial circulator velocity, initial volume conditions, and the homologous curves. This quantity must be nonzero if the relative circulator motor torque table is entered.
W9(R)	TF2 friction torque coefficient ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$). This parameter multiplies the speed ratio (absolute circulator speed/rated speed) to the second power. The friction torque factors are summed together.
W10(R)	TF0, friction torque coefficient ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$). This is constant frictional torque.
W11(R)	TF1, friction torque coefficient ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$). This multiplies the speed ratio to the first power.
W12(R)	TF3, friction torque coefficient. ($\text{N}\cdot\text{m}$, $\text{lb}_f\cdot\text{ft}$). This multiplies the speed ratio to the third power.

8.17.11 Circulator Variable Inertia, CCC0308

This card is optional, and if this card is not entered, the circulator inertia is given by Word 6 of card CCC0302. If this card is entered, circulator inertia is computed from

$$I = I_3 S^3 + I_2 S^2 + I_1 S^1 + I_0$$

where S is the relative circulator speed defined as the absolute value of the circulator rotational velocity divided by the rated rotational velocity.

W1(R) Relative speed at which to use the cubic expression for inertia. When the relative speed is less than this quantity, the inertia from Word 6 of card CCC0302 is used.

W2-W5(R) I_3, I_2, I_1, I_0 (kg/m², lb·ft²).

8.17.12 Circulator-Shaft Connection, CCC0309

This card is optional, and if this card is entered, the circulator is connected to a SHAFT component. The circulator may still be driven by a circulator motor that can be described in this component, by a turbine also connected to the SHAFT component, or from torque computed by the control system and applied to the SHAFT component. The circulator speed table may not be entered if this card is entered.

W1(I) Control component number of the shaft component.

W2(I) circulator disconnect trip. If this quantity is omitted or 0, the circulator is always connected to the SHAFT. If nonzero, the circulator is connected to the shaft when the trip is false and disconnected when the trip is true.

8.17.13 Relative circulator Motor Torque Data, CCC6001-6099

These cards are required only if W1 of card CCC0301 is 0. If the circulator velocity table is not being used and these cards are present, the torque-inertia equation is used. When the electrical power is supplied to the circulator motor (the circulator trip is off), the net torque is computed from the rated circulator motor torque times the relative circulator motor torque from this table and the torque from the homologous data. If the electrical power is disconnected from the circulator (the circulator trip is on), the circulator motor torque is 0.

W1(R) Circulator velocity (rad/s, rev/min).

W2(R) Relative circulator motor torque.

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs.

8.17.14 Time-Dependent circulator Velocity Control, CCC6100

This card is required only if W2 of card CCC0301 is 0. The velocity table, if present, has priority in setting the circulator velocity over the circulator trip, the circulator motor torque data, and the torque-inertia equation.

- W1(I) Trip number. If the trip number is 0, the circulator velocity is always computed from this table. If the trip number is nonzero, the trip determines which table is to be used. If the trip is off, the circulator velocity is set from the trip, the circulator motor torque data, and the torque-inertia equation as if this table had not been entered. If the trip is on, the circulator velocity is computed from this table. If Word 2 is missing, the search variable in the table is time and the search argument is time minus the trip time. If this word is used, it takes precedence over the trip number used in Word 6 of the CCC0301 card.
- W2(A) Alphanumeric part of variable request code. This quantity is optional. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. TIME can be selected, but the trip time is not subtracted from the advancement time.
- W3(I) Numeric part of variable request code. This is assumed to be 0 if missing.

8.17.15 Time-Dependent circulator Velocity, CCC6101-6199

These cards are required only if W2 of card CCC0301 is 0.

- W1(R) Search variable. Units depend on the quantity selected for the search variable.
- W2(R) Circulator velocity (rad/s, rev/min).

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs. Time values must be in increasing order.

8.17.16 Tabulated circulator pressure ratio data input, CCC7NNX

These cards are required for circulator. Pressure ratio data are tabulated by RPM and corrected mass flow rate. The quantity NN may be 01-10. X can be 1 to 9.

8.17.16.1 Number of data card, CCC7000

Enter the number of tabulated pressure ratio data sets. A data set is the corrected mass flow rate and pressure ratio pairs for a RPM value.

- W1(I) NumPrTBL. Number of pressure ratio data sets.

8.17.16.2 Pressure ratio data cards, CCC7NN1-NN9

NN can be 01 to 10. This number must be equal to the W1(I) of CCC70000 card.

W1(R) Enter the RPM number for the pressure ratio data set.

W2(R) Enter the corrected mass flow rate.

W3(R) Enter the pressure ratio.

W4(R)~W41(R) Additional corrected mass flow rate and pressure ratio pairs. Up to 20 pairs can be entered for one RPM set.

8.17.17 Tabulated circulator efficiency data input, CCC8NNX

These cards are required for circulator. Efficiency data are tabulated by RPM and corrected mass flow rate. The number NN may be 01-10. The quantity X can be 1 to 9.

8.17.17.1 Number of data card, CCC8000

Enter the number of tabulated efficiency data sets. A data set is the corrected mass flow rate and efficiency pairs for a RPM value.

W1(I) NumEffTBL. Number of efficiency data sets.

8.17.17.2 Efficiency data cards, CCC8NN1-NN9

NN can be 01 to 10. This number must be equal to the W1(I) of CCC80000 card.

W1(R) Enter the RPM number for the efficiency data set.

W2(R) Enter the corrected mass flow rate.

W3(R) Enter the efficiency.

W4(R)~W41(R) Additional corrected mass flow rate and efficiency pairs. Up to 20 pairs can be entered for one RPM set.

8.18 Multiple Junction Component, MTPLJUN

A multiple junction component is indicated by MTPLJUN for Word 2 on card CCC0000.

The one or more junctions specified by this component can connect volumes in the same manner as several single-junction components except that all the volumes connected by the junctions in the